

An Internal Force Transfers Energy; a Hinge Motor Can Also Supply It

Arm Mechanical Energy
Base Input $- F \cdot v_H - n \omega_{arm}$

Club Mechanical Energy
 $F \cdot v_H + n \omega_{club}$



Signed Hinge-Force Power: $F \cdot v_H$
Direction Depends on the State

Combined Input: Base Power $+ n (\omega_{club} - \omega_{arm})$
Gravity Is Included in Each Body's Potential; No Dissipation Is Modeled